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R&D Spending and Pharmaceutical Innovation:

A Cross-Country Panel Data Analysis

1. Introduction

Pharmaceutical innovation is among the most essential aspects of human welfare. It contributes to new treatments for infectious diseases, cancers, and other conditions that affect populations worldwide. Even so, the process of bringing a new drug from laboratory discovery to regulatory approval is extremely expensive and uncertain. Understanding what institutional and economic factors drive innovation activity is, therefore, a question of both academic and policy importance.

This study investigates the effect of research and development (R&D) spending on pharmaceutical innovation across countries over time. The research question guiding this analysis is: to what extent does R&D expenditure influence the number of clinical trials conducted within a country-year? Clinical trials serve as a meaningful proxy for pharmaceutical innovation because they represent the journey from research investment to active human-subject testing.

This topic is of broad economic and social importance. Pharmaceutical innovation contributes directly to medical advances, life expectancy, and overall public health. As far as policy is concerned, governments and international bodies seek to understand how research investment translates into innovation outputs so that spending decisions and incentive structures can be designed accordingly. Cross-country variation in both R&D investment and trial activity makes an international panel dataset appropriate for studying this relationship.

A key dimension of this topic is understanding who bears the cost of pharmaceutical R&D investment, and how large a share of total research spending is directed toward health. Globally, pharmaceutical R&D is funded predominantly by private business enterprises rather than governments. Across 26 OECD countries, external public and non-profit sources accounted for just under 10 percent of total pharmaceutical R&D spending in 2019, meaning that the private sector finances roughly 90 percent of industry research (OECD, 2023). The government share varies considerably by country: public subsidies covered only 1 to 2 percent of pharmaceutical R&D expenditure in Japan and Switzerland, but reached as high as 49 percent in the United Kingdom (OECD, 2023). In the United States, the federal government shapes private R&D activity indirectly through programs such as Medicare that expand the demand for prescription drugs, and through NIH-funded basic research that lowers the cost of early-stage discovery (Congressional Budget Office, 2021). With respect to the overall scale of health-directed research investment, global pharmaceutical R&D expenditure exceeded $300 billion in 2023, more than doubling from $137 billion in 2012 (Statista, 2024). The fact that private actors dominate pharmaceutical R&D spending, but do so at very different intensities across countries, is precisely what motivates the use of national R&D expenditure as a share of GDP as our primary explanatory variable. Because this measure captures both public and private research outlays, it reflects the overall research ecosystem of each country rather than any single funding source.

A large body of literature has studied the economics of R&D and pharmaceutical innovation. DiMasi, Grabowski, and Hansen (2016) provide influential estimates of the private costs of drug development, finding that the average capitalized cost of bringing a new compound to market exceeds $2.5 billion. Moreover, clinical trials represent approximately 50 to 58 percent of total R&D expenditure (Simoens and Huys, 2021). This highlights clinical trial activity as both a result of R&D investment and a major cost center.

The demand-side drivers of innovation are well-explained by Acemoglu and Linn (2004). They provide a foundational theoretical and empirical contribution. Using exogenous variation in potential market size driven by U.S. demographic trends, they find that a one percent increase in potential market size leads to a four to six percent increase in new drug entry within a therapeutic category. Their results establish that pharmaceutical R&D responds strongly to expected revenues. Therefore, we have decided to include GDP per capita and population in our model as proxies for market size and capacity to pay.

Toole (2012), building on the Acemoglu-Linn framework, introduces public research investment as an additional supply-side determinant, finding that NIH-funded basic research has economically and statistically significant effects on new drug entry, particularly at the discovery stage. This finding is relevant to our model because national R&D expenditure as a share of GDP captures both public and private research outlays, making it a broader measure of research ecosystem investment than firm-level R&D alone.

The Congressional Budget Office (2021) documents that pharmaceutical R&D spending in the United States has grown approximately tenfold since the 1980s in real terms, while clinical trial success rates have declined. This inverse trend motivates the use of trial counts rather than drug approvals as our dependent variable, since approvals may reflect regulatory lags and trial success rates in addition to innovation effort.

This project contributes to the literature by examining country-year panel data from twenty randomly selected countries over the period 2010 to 2020, combining clinical trial data from ClinicalTrials.gov with macroeconomic indicators from the World Bank World Development Indicators. By controlling for GDP per capita, population, and health expenditure, the analysis aims to isolate the partial effect of R&D spending on pharmaceutical innovation activity beyond what is explained by overall economic development.

2. Model Specification

The empirical approach in this study is based on a multiple linear regression (MLR) framework applied to a country-year panel dataset. The dependent variable, TrialCount, measures the number of pharmaceutical clinical trials conducted in each country in a given year and serves as a proxy for pharmaceutical innovation activity.

The econometric model is specified as follows:

$$ln(TrialCount_{it} + 1) = \beta_0 + \beta_1 RDSpend_{it} + \beta_2 ln(GDPpc_{it}) + \beta_3 ln(Pop_{it}) + \beta_4 HSpend_{it} + u_{it}$$

where *i* indexes country and *t* indexes year. The variables are defined as follows:

- $ln(TrialCount)_{it}$ = log of the number of pharmaceutical clinical trials conducted in country *i* in year *t*, plus one (dependent variable). The log transformation is applied because TrialCount is highly right-skewed.
- $RDSpend_{it}$ = research and development expenditure as a percentage of GDP in country i in year t (primary explanatory variable).
- $ln(GDPpc)_{it}$ = log of GDP per capita in current U.S. dollars in country i in year t.
- $ln(Pop)_{it}$ = log of total population of country i in year t.
- $HSpend_{it}$ = current health expenditure as a percentage of GDP in country i in year t.
- u_{it} = the error term.

The expected signs for each parameter are as follows:

$\beta_1 > 0$: R&D spending is the variable of primary interest. Countries that invest more in research as a share of their economy should have a stronger research ecosystem and generate more clinical trial activity (Acemoglu and Linn, 2004; Toole, 2012).

$\beta_2 > 0$: Wealthier countries typically possess more developed research institutions, better regulatory infrastructure, and greater capacity to attract pharmaceutical firms and trial sponsors.

$\beta_3 > 0$: Larger populations represent a larger potential patient pool for clinical trials, reducing recruitment costs. Market size, which is partly a function of population, is a strong positive predictor of drug development activity (Acemoglu and Linn, 2004).

$\beta_4 > 0$: Countries with higher health expenditure as a share of GDP have stronger healthcare infrastructure -- including hospital networks, trained clinicians, and regulatory capacity -- that facilitates the conduct of clinical trials.

3. Data

3.1 Sample Selection and Data Sources

To construct the dataset for this analysis, twenty countries were randomly selected to avoid any bias. The sample spans low-income, lower-middle-income, upper-middle-income, and high-income economies drawn from Sub-Saharan Africa, South Asia, East Asia, the Pacific, the Middle East, North Africa, Latin America, Eastern Europe, and Western Europe. Table 1 lists the twenty countries included in the sample.

Table 1 Sample Countries (N = 20)

Country	Income Group | Region	Country	Income Group | Region
Brazil	Upper-middle income | Latin America	Canada	High income | North America
Egypt	Lower-middle income | N. Africa & Middle East	Germany	High income | Western Europe
Ghana	Lower-middle income | Sub-Saharan Africa	India	Lower-middle income | South Asia
Indonesia	Upper-middle income | East Asia & Pacific	Israel	High income | Middle East
Japan	High income | East Asia & Pacific	Kenya	Low income | Sub-Saharan Africa
Mexico	Upper-middle income | Latin America	Morocco	Lower-middle income | N. Africa & Middle East
Netherlands	High income | Western Europe	Nigeria	Low income | Sub-Saharan Africa
Pakistan	Lower-middle income | South Asia	Poland	High income | Eastern Europe
South Africa	Upper-middle income | Sub-Saharan Africa	South Korea	High income | East Asia & Pacific
Turkey	Upper-middle income | Middle East & Europe	United Kingdom	High income | Western Europe

Note: Income classifications follow the World Bank Atlas method.
Source: World Bank Country and Lending Groups classification.

The panel covers the period 2010 to 2020, yielding 220 country-year observations (20 countries x 11 years). The dependent variable, TrialCount, is the number of pharmaceutical clinical trials registered per country per year, sourced from ClinicalTrials.gov. The four independent variables are sourced from the World Bank World Development Indicators (WDI) database, downloaded in April 2026.

An important data quality note concerns the R&D spending variable. The six countries with incomplete R&D data were handled using two different approaches depending on how many observations were available. For Indonesia, which had six of eleven years available, and Pakistan, which had five of eleven years, linear interpolation was applied. Because both countries had multiple data points spanning the period, interpolation is methodologically appropriate -- it estimates missing values by drawing a line between two known observations. For Ghana, Kenya, Morocco, and Nigeria, however, only a single year of R&D data was reported to the World Bank over the entire 2010-2020 period. With only one data point, linear interpolation is not possible, as it requires at least two known values. Instead, a constant fill was applied: the single available observation was carried forward and backward across all eleven years, treating R&D spending as unchanged throughout the period. This is a stronger assumption and introduces greater measurement error for these four countries. It is, however, a defensible approximation given that low-income countries with minimal R&D infrastructure tend to exhibit very stable, low levels of research expenditure over time. This approach maintains a balanced panel of 220 observations. Two robustness checks are reported: (1) excluding all six countries with filled R&D values entirely, and (2) zero-filling all missing R&D observations -- treating them as zero expenditure.

3.2 Summary Statistics

Table 2 presents descriptive statistics for all five variables used in the econometric model. For each variable, the table reports the mean, standard deviation, minimum, and maximum values across all 220 country-year observations.

Table 2 Descriptive Statistics

Variable	Mean (SD)	Min	Max	Source
TrialCount (trials/country-year)	511.0 (495.0)	2	1,785	ClinicalTrials.gov
RDSpend (% of GDP)	1.43 (1.28)	0.08	5.83	World Bank WDI
GDPpc (thousands of current USD)	18.45 (18.56)	0.99	54.70	World Bank WDI
Population (millions)	154.3 (278.9)	7.62	1,402.6	World Bank WDI
Health expenditure (% of GDP)	6.52 (2.87)	2.22	13.02	World Bank WDI

Note: Standard deviations in parentheses. GDPpc is reported in thousands of current USD. Population is reported in millions. RDSpend for Ghana, Kenya, Morocco, and Nigeria was filled by carrying a single observed value forward and backward across all years (constant fill); Indonesia and Pakistan were handled via linear interpolation across multiple observed values. N = 220 country-year observations (20 countries, 2010-2020).
Source: ClinicalTrials.gov (U.S. National Library of Medicine); World Bank World Development Indicators (April 2026 download).

The dependent variable TrialCount ranges from a minimum of 2 to a maximum of 1,785 trials per country-year, with a mean of 511 and a standard deviation of 495. The large standard deviation relative to the mean confirms the highly right-skewed distribution of trial activity, driven by a small number of high-income nations. This skewness motivates the log transformation ln(TrialCount + 1) applied in the regression. R&D spending ranges from 0.08 to 5.83 percent of GDP with a mean of 1.43 percent, reflecting the gap between low-income countries such as Nigeria and Ghana and innovation leaders such as Israel. GDP per capita averages $18,450 but has a standard deviation nearly equal to the mean, reflecting the broad income dispersion across the sample. Health expenditure averages 6.52 percent of GDP. Population averages 154.3 million but is heavily influenced by India, whose 1.4 billion people are far larger than any other country in the sample.

3.3 Correlation Analysis

Table 3 presents the correlation matrix for all five variables, computed from the 220-observation merged panel.

Table 3 Correlation Matrix

Variable	(1)	(2)	(3)	(4)	(5)
(1) TrialCount	1.000				
(2) RDSpend	0.588	1.000			
(3) GDPpc	0.749	0.784	1.000		
(4) Population	-0.184	-0.230	-0.302	1.000	
(5) HSpend	0.690	0.614	0.851	-0.341	1.000

Note: All correlations computed on the full 220 country-year observations. Values rounded to three decimal places. Source: ClinicalTrials.gov; World Bank World Development Indicators (April 2026 download).

The correlation between RDSpend and TrialCount is 0.588, providing preliminary support for the main hypothesis. GDP per capita has the strongest bivariate relationship with TrialCount (r = 0.749), reflecting the important role of economic development in driving research. Health expenditure is also strongly positively correlated with TrialCount (r = 0.690). Population is negatively correlated with TrialCount (r = -0.184), which is surprising but is explained by the composition of the sample: the most populous countries (India, Nigeria, Pakistan, and Indonesia) are also among the lowest-income and conduct comparatively fewer trials. Once income is controlled for in the multiple regression, the partial effect of population may turn out to be positive.

Among the explanatory variables, the strongest correlation is between GDPpc and HSpend (r = 0.851). This raises a multicollinearity concern that is addressed in the regression analysis.

RDSpend and GDPpc are also moderately correlated (r = 0.784), which is expected given that richer countries have greater capacity to fund research.

3.4 Key Data Considerations

The dataset is a country-year panel combining cross-sectional and time-series variation across 20 countries and 11 years. Both TrialCount and Population are highly right-skewed, motivating log transformations: ln(TrialCount + 1) for the dependent variable and ln(Population) for the regressor. GDP per capita is also log-transformed. The potential for reverse causality between TrialCount and RDSpend -- where countries with greater trial activity may justify higher R&D budgets -- is acknowledged as a limitation that cannot be fully addressed with the current data.

4. Results

Table 4 presents the estimation results from three regression models. Model 1 is the main specification: a pooled OLS regression of ln(TrialCount + 1) on RDSpend, ln(GDPpc), ln(Population), and HSpend, estimated on the full 220-observation panel. Missing R&D values for Indonesia and Pakistan were filled via linear interpolation, while Ghana, Kenya, Morocco, and Nigeria were handled using a constant fill from a single observed value. Model 2 is a robustness check that excludes all six countries with filled R&D data entirely, leaving 154 observations across 14 countries. Model 3 replaces all missing R&D values with zero rather than imputing them.

Table 4 Regression Results: R&D Spending and Clinical Trial Activity

	Main (Imputed) (1)	**Excl. Imputed (2)**	**Zero-Fill (3)**
Dependent variable: ln(TrialCount + 1)			
RD Spend (% GDP)	0.129	0.049	0.251***
	(0.082)	(0.048)	(0.081)
ln(GDP per capita)	1.067***	0.544***	0.918***
	(0.120)	(0.103)	(0.127)
ln(Population)	0.519***	0.154***	0.510***
	(0.068)	(0.054)	(0.067)
Health Spend (% GDP)	0.005	-0.018	0.020
	(0.044)	(0.029)	(0.043)
Constant	-13.972***	-1.677	-12.693***
	(1.630)	(1.591)	(1.675)
Observations	220	154	220
R-squared	0.676	0.470	0.686
Adjusted R-squared	0.670	0.456	0.680

*Note: Standard errors in parentheses. ***p < 0.01, **p < 0.05, *p < 0.1. Model 1 uses linear interpolation for Indonesia and Pakistan, and constant fill for Ghana, Kenya, Morocco, and Nigeria. Model 2 excludes all six countries with filled R&D data. Model 3 replaces missing R&D values with zero.*

The main result in Model 1 does not support the central hypothesis. The coefficient on RDSpend is 0.129 with a standard error of 0.082, which is not statistically significant at any conventional level ($p > 0.1$). While the positive sign is consistent with the expected direction, the estimate cannot be distinguished from zero. This means that, after controlling for GDP per capita, population, and health expenditure, there is no statistically reliable evidence that higher R&D spending as a share of GDP is associated with greater clinical trial activity in this sample.

The control variables, by contrast, perform largely as expected. The coefficient on ln(GDPpc) is 1.067 and is highly significant ($p < 0.01$), indicating that a one percent increase in GDP per capita is associated with approximately a one percent increase in trial activity. This is consistent with Acemoglu and Linn (2004), who show that wealthier markets attract substantially more pharmaceutical R&D. The coefficient on ln(Population) is 0.519 and is also highly significant ($p < 0.01$), confirming that larger countries host more trials once income is held constant. Notably, the sign is positive in the multivariate regression, whereas the raw correlation between population and trial count was negative ($r = -0.184$) -- exactly the reversal anticipated in Section 3. Health expenditure (HSpend) carries a small positive coefficient of 0.005 but is not statistically significant ($p > 0.1$), suggesting that health system spending does not independently predict trial activity once economic development is accounted for.

The overall model fit is moderate. The R-squared is 0.676, indicating that the four regressors together explain approximately 68 percent of the variation in log trial counts across the 220 observations. The F-statistic of 112.1 ($p < 0.001$) confirms that the model as a whole is statistically significant, driven primarily by the strong explanatory power of GDP per capita and population. The elevated VIF on ln(GDPpc) (6.22) is consistent with the high correlation between GDPpc and HSpend observed in Table 3 ($r = 0.851$), but remains below the threshold of 10 that would indicate severe multicollinearity.

The two robustness checks reveal an important sensitivity of the RDSpend estimate to how missing data is handled. In Model 2, which excludes the six imputed countries entirely, the RDSpend coefficient falls further to 0.049 and remains insignificant. The R-squared also drops substantially to 0.470, and the GDPpc coefficient nearly halves to 0.544. This pattern suggests that much of the explanatory power in Model 1 is driven by the imputed countries, which span a wide range of income levels and trial activity. In Model 3, where missing R&D values are replaced with zero, the RDSpend coefficient rises to 0.251 and becomes statistically significant at the one percent level. This result warrants careful interpretation. Zero-filling creates a sharp contrast between the imputed countries (assigned RDSpend = 0) and the rest of the sample, which have positive and often substantial R&D values. This amplified variation mechanically strengthens the estimated relationship between R&D spending and trial counts, which is why the coefficient is larger and more significant than in Model 1. In other words, the significance in Model 3 may reflect the mechanical effect of the imputation assumption rather than a genuine underlying relationship.

Taken together, the three models tell a consistent story: economic development, as measured by GDP per capita and population, is the dominant predictor of clinical trial activity in this cross-country sample. R&D spending shows a positive but statistically weak relationship in the main specification, and its significance emerges only under the most aggressive imputation assumption. This does not necessarily mean that R&D investment does not matter for pharmaceutical innovation. Rather, it may reflect that the relationship is better identified using firm-level or sector-specific R&D data, or that a larger and more complete cross-country dataset is needed to detect the effect reliably.

5. Summary and Conclusion

This paper examined the effect of research and development expenditure on pharmaceutical innovation, measured by the number of clinical trials conducted per country per year, using a cross-country panel dataset of 20 countries over the period 2010 to 2020. The primary explanatory variable was R&D spending as a percentage of GDP, with GDP per capita, population, and health expenditure included as controls. The analysis was motivated by a large body of literature -- including Acemoglu and Linn (2004), Toole (2012), and DiMasi et al. (2016) -- establishing that pharmaceutical innovation is driven by both the scale of research investment and the economic conditions that determine the expected returns to that investment.

The main finding is that R&D spending does not have a statistically significant effect on clinical trial counts in the primary specification. The coefficient on RDSpend is positive (0.129) but cannot be distinguished from zero at any conventional significance level. GDP per capita and population, by contrast, are both highly significant and positive, confirming that wealthier and larger countries host substantially more clinical trial activity. The positive sign on population in the multivariate regression -- reversing the negative raw correlation -- is consistent with theoretical expectations and underscores the importance of controlling for income when studying trial activity across countries with very different development levels. Health expenditure is not significant in any model, suggesting that, once GDP per capita is included, it does not independently explain variation in trial counts.

The robustness checks reveal that the null result for R&D spending is sensitive to data treatment choices. When the six countries with imputed R&D values are excluded, the coefficient falls further and remains insignificant. When missing values are zero-filled, the coefficient rises to 0.251 and becomes significant at the one percent level -- but this likely reflects a mechanical amplification of variation rather than a genuine effect. The consistency of the null result across the two more conservative specifications supports the conclusion that the evidence for an R&D spending effect is weak in this dataset.

Several limitations constrain the interpretation of these findings. First, the R&D data for six countries required imputation, introducing measurement error. Second, the sample of 20 countries and 11 years may not provide sufficient variation to detect an effect that operates over longer time horizons or through channels specific to high-income innovation ecosystems. Third, reverse causality remains a concern: countries with greater clinical trial activity may attract or justify higher R&D investment, biasing the OLS estimates. Fourth, the national R&D spending measure captures investment across all sectors, not pharmaceutical R&D specifically, which may further attenuate the estimated relationship.

Future research should consider sector-specific R&D data, a broader country sample, and instrumental variable approaches to address endogeneity. Despite the null result on R&D spending, this study contributes a carefully constructed cross-country panel dataset and a transparent econometric analysis that demonstrates the dominant role of economic development in explaining global variation in clinical trial activity.

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